

Accelerated Geometry

1st Semester

Unit 1 – Chap. 1(Basics of Geometry) & Chap. 2(Reasoning & Proofs)

- Points, Lines, and Planes
- Midpoint and Distance Formulas
- Perimeter & Area in the Coordinate Plane
- Measuring and Constructing Angles
- Inductive and Deductive Reasoning
- Algebraic Reasoning
- Proving Statements about Segments and Angles

Unit 2 – Chapter 3 (Parallel and Perpendicular Lines)

- Pairs of Lines and Angles
- Proofs with Parallel Lines and Perpendicular Lines
- Equations of Parallel and Perpendicular Lines

Unit 3 – Chapter 4 (Transformations)

- Translations
- Reflections
- Rotations
- Dilations

Unit 4 – Chapter 5 (Congruent Triangles) & Sections 6-1, 6-3, 6-5, & 6-6

- Angles of Triangles
- Congruent Polygons
- Proving Triangle Congruence (SSS, SAS, ASA, AAS, HL)
- Equilateral and Isosceles Triangles
- Coordinate Proofs
- Perpendicular Bisectors & Angle Bisectors
- Medians and Altitudes of Triangles
- Indirect Proof
- Inequalities in One or Two Triangles

Unit 5 – Chapter 7 (Quadrilaterals & Other Polygons)

- Angles of Polygons
- Parallelograms (properties, proving, special parallelograms)
- Properties of Trapezoids and Kites

2nd Semester

Unit 1 – Chapter 8 (Similarity)

- Similar Polygons
- Proving Triangle Similarity (AA~, SSS~, SAS~)
- Proportionality Theorems
- Triangle Midsegment Theorem

Unit 2 – Chap. 9(Right Triangles & Trigonometry)

- Pythagorean Theorem
- Special Right Triangles
- Similar Right Triangles
- Unit Circle
- Trigonometric Ratios (all 6 functions)
- Laws of Sine and Cosine
- Using both Radians and Degrees

Unit 3 – Chapter 10 (Circles)

- Lines and Segments that Intersect Circles
- Finding Arc Measures
- Inscribed Angles
- Angle Relationships
- Segment Relationships
- Circles in the Coordinate Plane
- Standard Form - Equation of a Circle

Unit 4 – Chapter 11 (Circumference, Area, and Volume)

- Circumference and Arc Length
- Area of Circles and Sectors
- Area of Polygons
- Volumes of Solids (Prism, Cylinder, Pyramid)
- Surface Area & Volume: Cones & Spheres

Unit 5 – Chapter 12 (Probability)

- Sample Space and Probability
- Independent and Dependent Events
- Two-Way Tables
- Disjoint and Overlapping Events
- Permutations and Combinations
- Binomial Distributions

Unit 6 – Parabolas (Algebra 2)

- state domain & range in interval notation
- Focus, Directrix, Focal diameter
- Converting equations to Standard Form and Vertex Form
- Completing the Square to convert to Standard Form
- Using Quadratic Formula for Real-Life problems